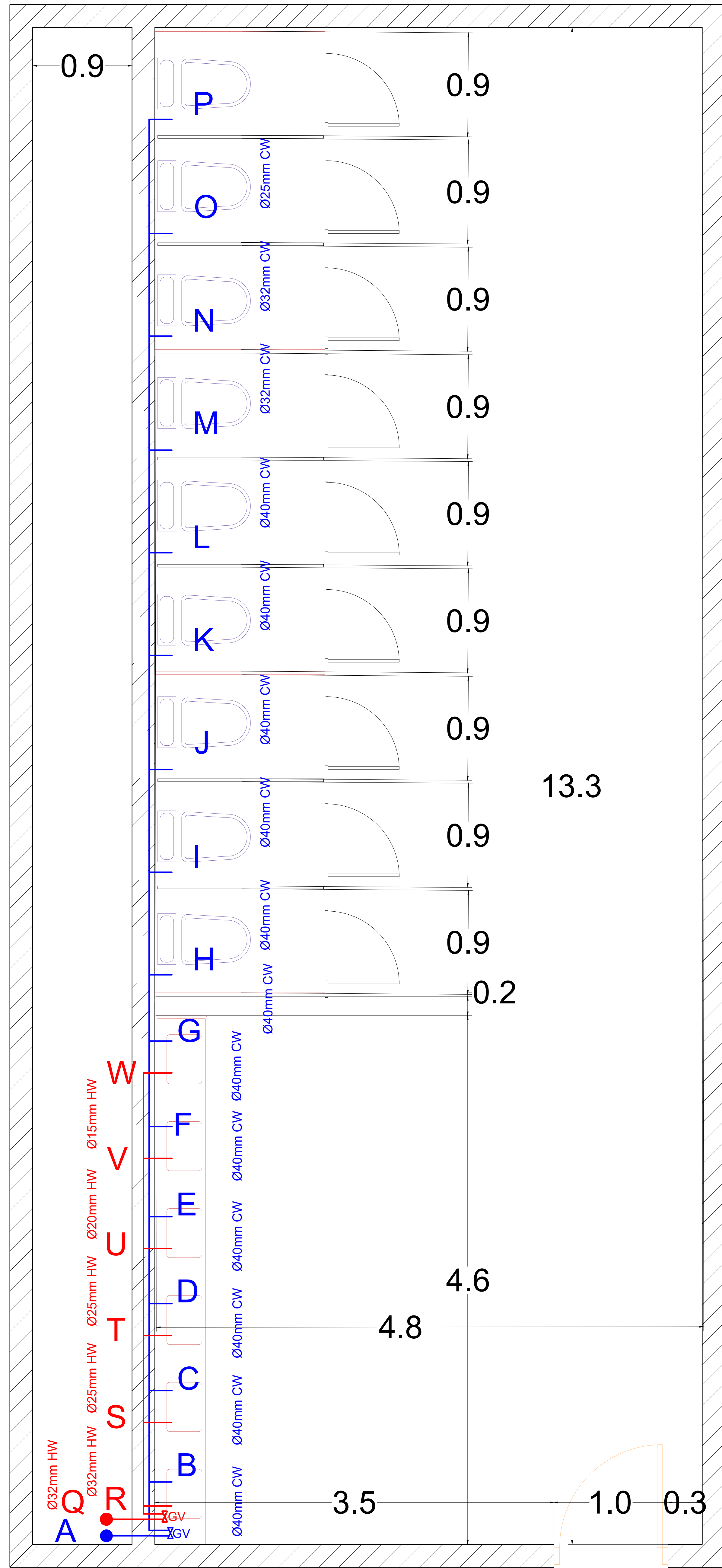




HOT WATER / COLD WATER SUPPLY LAYOUT



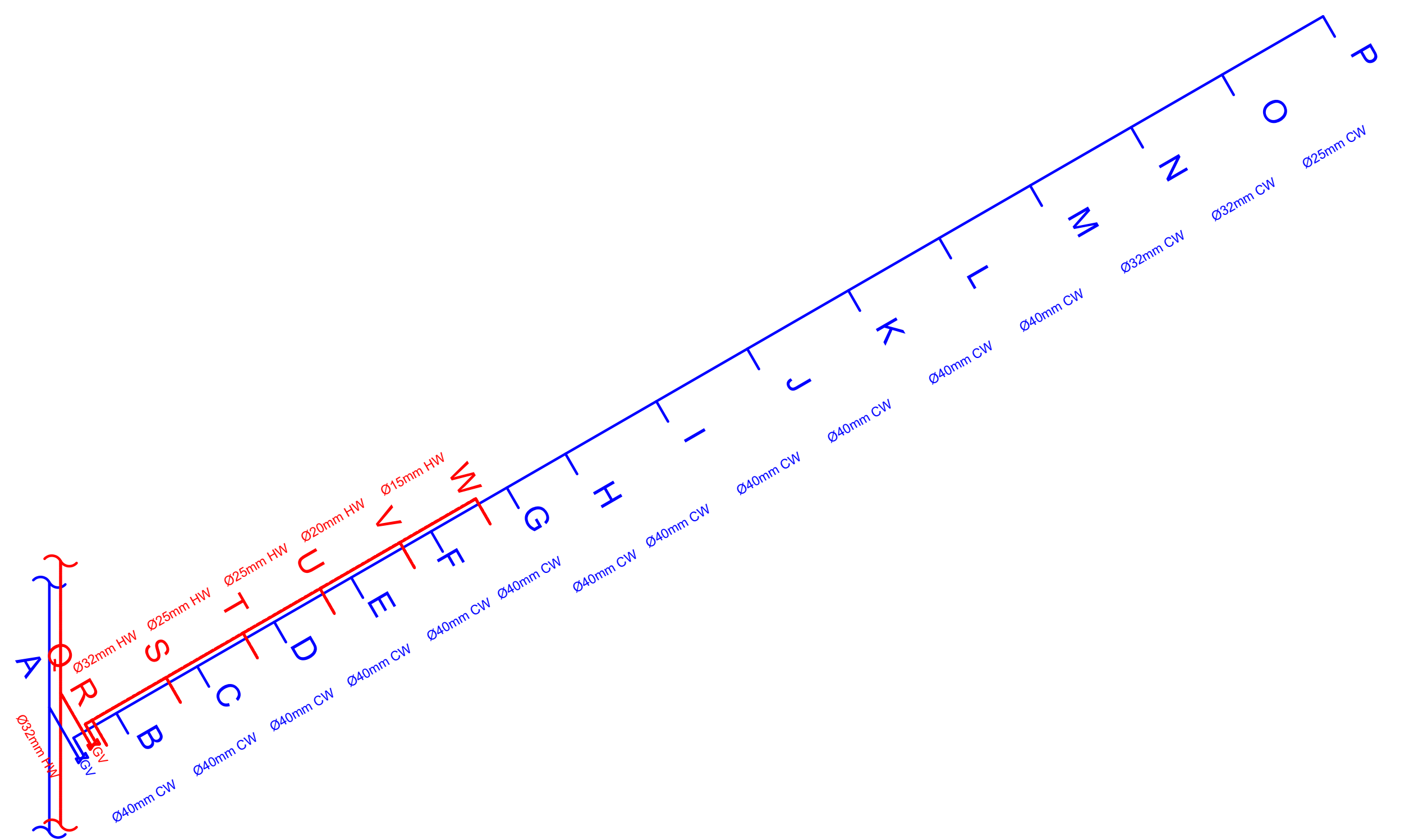
SANITARY DRAINAGE LAYOUT

Pipe	Fixtures Attached	Total Fixture Units	Flow Q (gpm)	Flow Q (m3/s)	Calculated Diameter	Selected Diameter
PO	W/C	5	9.4	0.0005922	0.022420414	25mm
ON	W/C X2	10	14.6	0.0009198	0.027941913	32mm
NM	W/C X3	15	17.5	0.0011025	0.030591356	32mm
ML	W/C X4	20	19.6	0.0012348	0.032374848	40mm
LK	W/C X5	25	21.5	0.0013545	0.033907747	40mm
KJ	W/C X6	30	23.3	0.0014679	0.035298615	40mm
JI	W/C X7	35	24.9	0.0015687	0.036490463	40mm
IH	W/C X8	40	26.3	0.0016569	0.037502272	40mm
HG	W/C X9	45	27.7	0.0017451	0.038487149	40mm
GF	W/C X9 + LAVATORY	46.5	28.12	0.00177156	0.038778175	40mm
FE	W/C X9 + LAVATORY X2	48	28.54	0.00179802	0.039066897	40mm
ED	W/C X9 + LAVATORY X3	49.5	28.96	0.00182448	0.039353103	40mm
DC	W/C X9 + LAVATORY X4	51	29.39	0.00185157	0.039644185	40mm
CB	W/C X9 + LAVATORY X5	52.5	29.825	0.001878975	0.039936494	40mm
BA	W/C X9 + LAVATORY X6	54	30.26	0.00190638	0.040229678	40mm

Calculation for pipe sizing for Cold Water

Pipe	Fixtures Attached	Total Fixture Units	Flow Q (gpm)	Flow Q (m3/s)	Calculated Diameter	Selected Diameter
WV	LAVATORY	1.5	4	0.000252	0.014625486	15mm
VU	LAVATORY X2	3	6.5	0.0004095	0.019843884	20mm
UT	LAVATORY X3	4.5	8.7	0.0005481	0.021569463	25mm
TS	LAVATORY X4	6	10.7	0.0006741	0.023920574	25mm
SR	LAVATORY X5	7.5	12.3	0.0007749	0.025646745	32mm
RQ	LAVATORY X6	9	13.7	0.0008631	0.027069992	32mm

Calculation for pipe sizing for Hot Water



HOT WATER / COLD WATER SUPPLY ISOMETRIC

Water demand for the building =
(20,000 Visitors x 5)+(100 staff x 15)= 100,500 gallons/day
100,500* 0.00378
=397.9 m³/day

Required storage = 2days

Volume required for underground square tank
= 397.9 * 2 =759.8 m³

Assumed water depth =4m

area required =759.8 / 4 = 189.8 m²

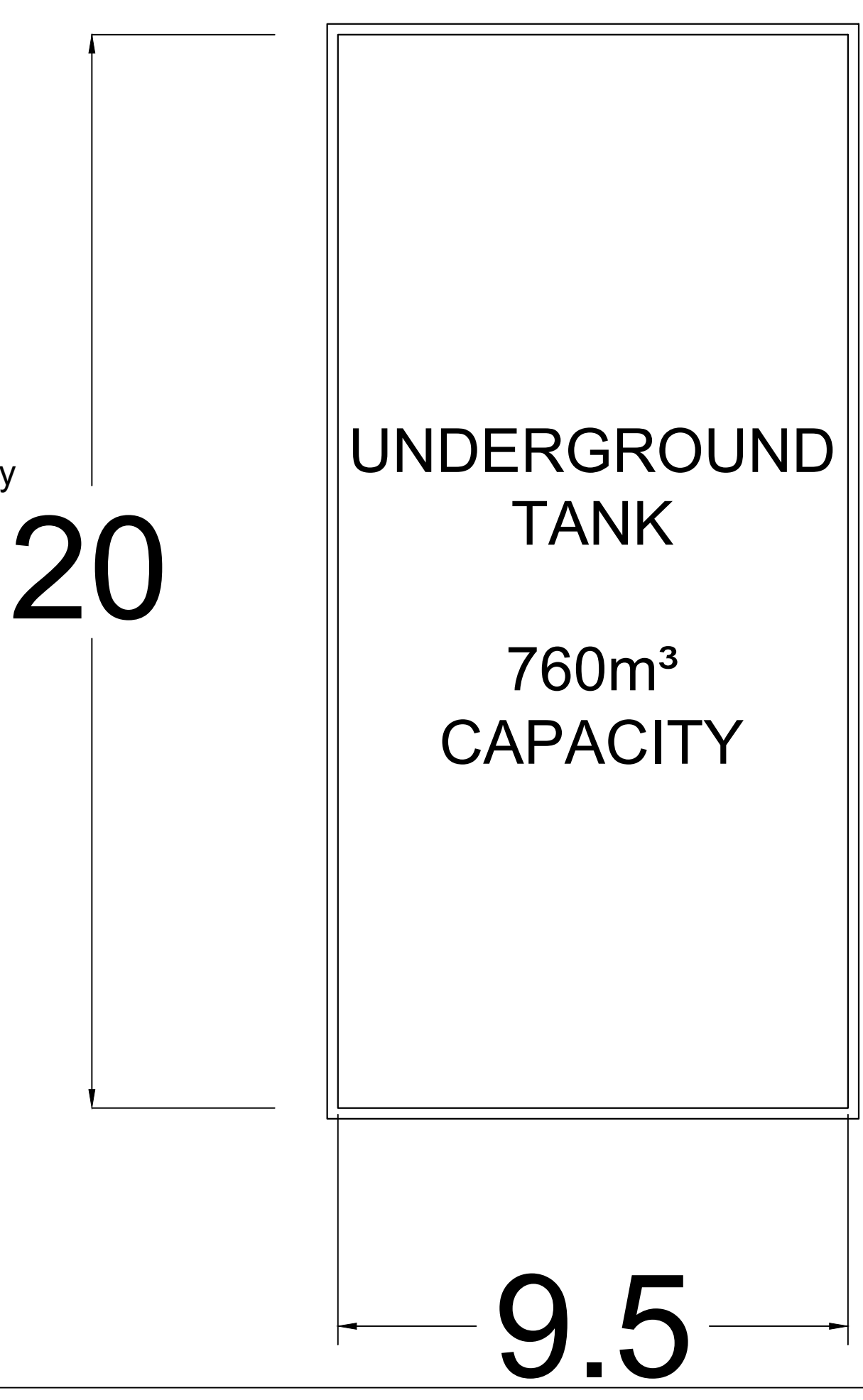
20m x 9.5m= 190m² underground water tank

water depth :4m

free board : 0.25m

total depth : 4.25m

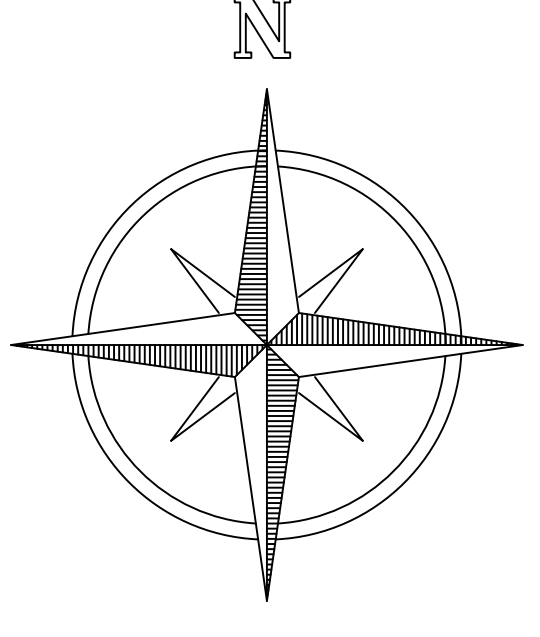
volume of water : 760 m³





جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER

SYMBOLS:

- COLD WATER
- HOT WATER
- GATE VALVE
- WASTE PIPE
- VENT PIPE
- SOIL PIPE
- MFT

Design Studio (7), Technical Project, 2024/2025

INSTRUCTORS

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DONE BY:
ABDULELAH ALQABBANI

PROJECT:
MUSEUM

DRAWING NAME:
PLUMBING DETAILS

SCALE:
VARIES

SHEET NUMBER:
A21